THE TRUTH ABOUT NIGERIA'S POWER CRISIS

A Data-Driven Analysis of 2023-2024 Generation Trends

AltSchool Africa Data Science Capstone Project

Date: February 17, 2026

Analysis Period: January 2023 to December 2024 (731 days)

Executive Summary

Here is the short version for anyone too busy to read the full thing: Nigeria's power generation declined 8.6% in 2024. The system became 41% more unpredictable. The deterioration was statistically significant, practically meaningful, and accelerating by year-end.

That is not a rough stretch or an unlucky quarter. It is a measurable, documented slide backed by 731 days of data and confirmed by independent statistical tests.

The longer version, with the evidence, follows below.

What Happened

The Headline Numbers

Before we look at any chart, here are the numbers that frame everything else:

Metric	2023	2024	Change
Mean Daily Generation	4,018.6 MWh	3,674.4 MWh	-344.2 MWh (-8.6%)
Std Deviation	451.6 MWh	580.5 MWh	+128.9 (+28.5%)
Coefficient of Variation	11.2%	15.8%	+4.6pp (+41%)
Frequency Violations	156 days (42.7%)	179 days (48.9%)	+23 days
Statistical Anomalies	2	11	+9 (5.5x more)

The mean dropped. The variability rose. The technical indicators worsened. Every angle of this analysis points in the same direction.

Figure 1: Daily Power Generation, 2023 vs 2024

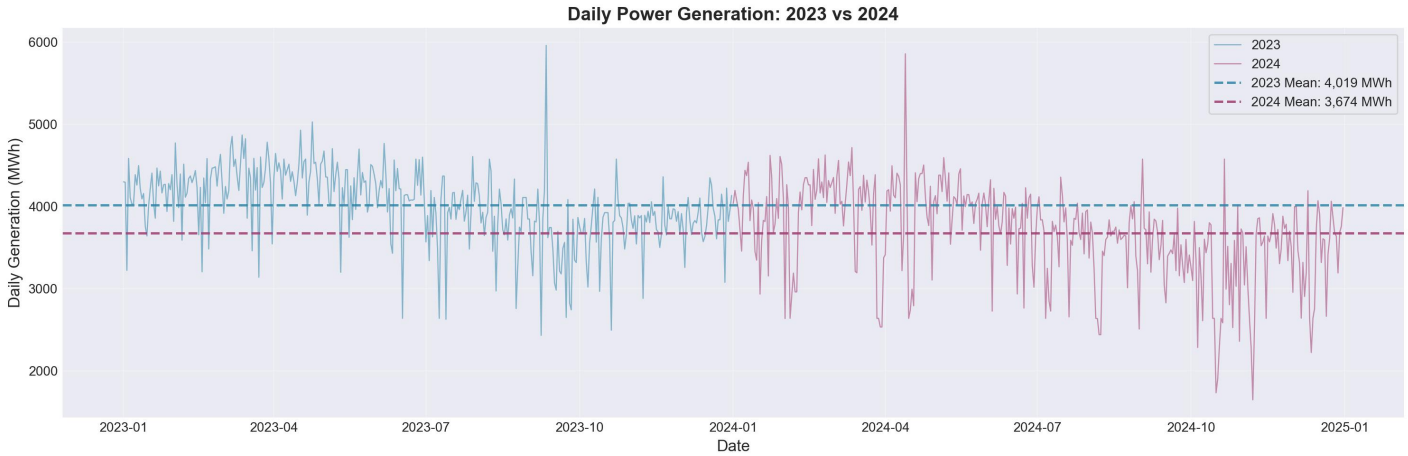

fig_01_daily_generation_timeseries.png

This is the raw view. Each line is one year of daily output plotted over time. The dashed horizontal lines are the annual averages.

What to notice: the 2024 line sits visibly lower than the 2023 line through most of the year. It also swings wider, especially from mid-year onward. The gap between the two dashed average lines is **344 MWh**. That is roughly what a small city uses in a day, gone from the national grid, every single day of 2024.

The raw data is noisy by nature. Power systems have good days and bad days. But even through that noise, the separation between years is visible. To confirm the direction of travel, we smooth it out.

Figure 2: 30-Day Rolling Average

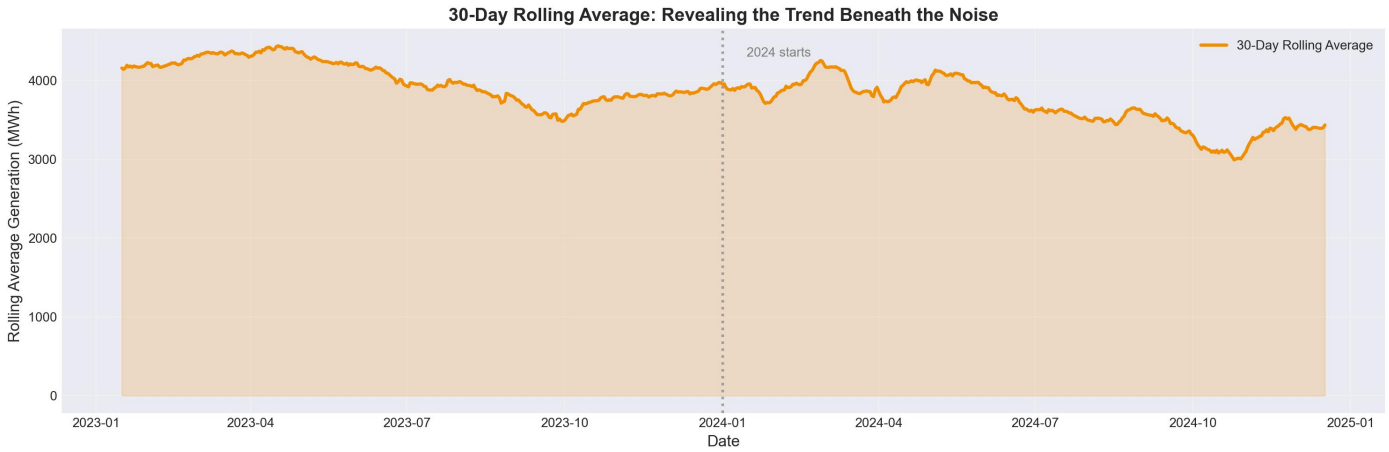

fig_02_rolling_average_trend.png

A 30-day rolling average suppresses day-to-day noise and shows the underlying trend. Think of it as asking: *ignoring today's specific circumstances, what direction is the system actually moving?*

The answer is down. The curve rises through early 2023, peaks around April-May 2023, and then declines with increasing steepness through 2024. The drop accelerates noticeably from mid-2024 onward, especially in Q4.

The vertical dotted line marks where 2024 begins. Everything to its right is 2024, and the slope after that line is steeper than anything we saw in 2023. This chart is what motivated the statistical tests in the next section. We saw the pattern; now we confirm it rigorously.

Why We Can Be Confident This is Real

A 344 MWh difference between two years could theoretically be random variation. Two consecutive years of anything will not be identical. So the question is whether this decline is statistically meaningful or just noise.

The T-Test

We ran a two-sample t-test comparing all 365 days of 2023 against all 366 days of 2024.

- t-statistic: 8.9469
- p-value: 0.00000005 (essentially zero)
- Effect size (Cohen's *d*): 0.6618

The p-value being near zero means the probability of observing a difference this large due to chance alone is negligible. We reject the null hypothesis. The decline is statistically real.

Cohen's *d* of 0.66 is classified as a "large" effect. This matters because statistical significance alone does not tell us whether the difference is big enough to care about in practice. A large effect size tells us it is.

Figure 3: Distributions and Confidence Intervals

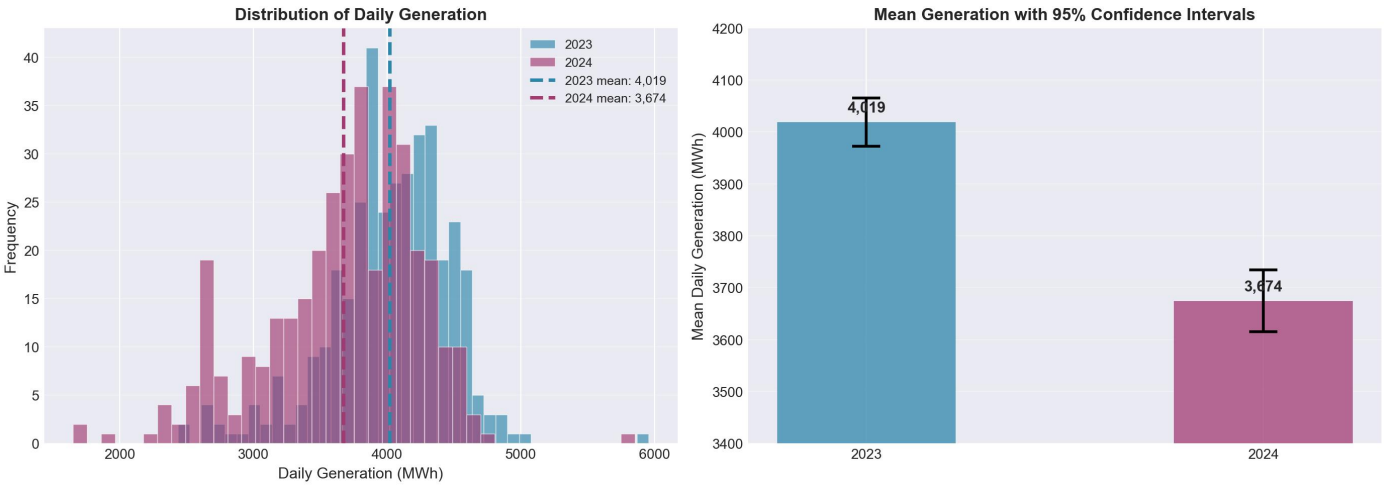

fig_03_distributions_and_ci.png

Two panels here. On the left, the overlapping histograms show the shape of each year's daily generation values. The 2024 histogram is shifted left compared to 2023, confirming lower generation. It is also wider and shorter, confirming higher variability. When a distribution widens, it means more days deviated sharply from the average in both directions. That is what instability looks like in data form.

On the right, the confidence interval chart gives a different view of the same result. Each bar is a year's mean generation, and the error bars represent the 95% confidence interval around that mean. The two intervals do not come close to overlapping:

- 2023 CI: [3,972 MWh, 4,065 MWh]
- 2024 CI: [3,615 MWh, 3,734 MWh]

There is a gap of roughly 240 MWh between the upper bound of 2024's interval and the lower bound of 2023's. No matter how we adjust for sampling uncertainty, 2024's true mean falls well below 2023's. The statistics confirm what Figures 1 and 2 showed visually.

How Fast Was the Decline

Knowing the gap between years is useful. Knowing whether that gap widened or narrowed through the year is more useful for understanding where things are heading.

We grouped data into monthly averages and fitted linear regression lines to each year separately.

- 2023 monthly trend: -58.9 MWh per month ($R^2 = 0.63$)
- 2024 monthly trend: -72.2 MWh per month ($R^2 = 0.74$)

Both years declined. 2024 declined faster. The higher R^2 in 2024 means the downward trend was more consistent, less interrupted by recoveries. When a trend has a high R^2, it means the line fits the data tightly. The data in 2024 followed its downward path with grim consistency.

Figure 4: Monthly Averages with Regression Trend Lines

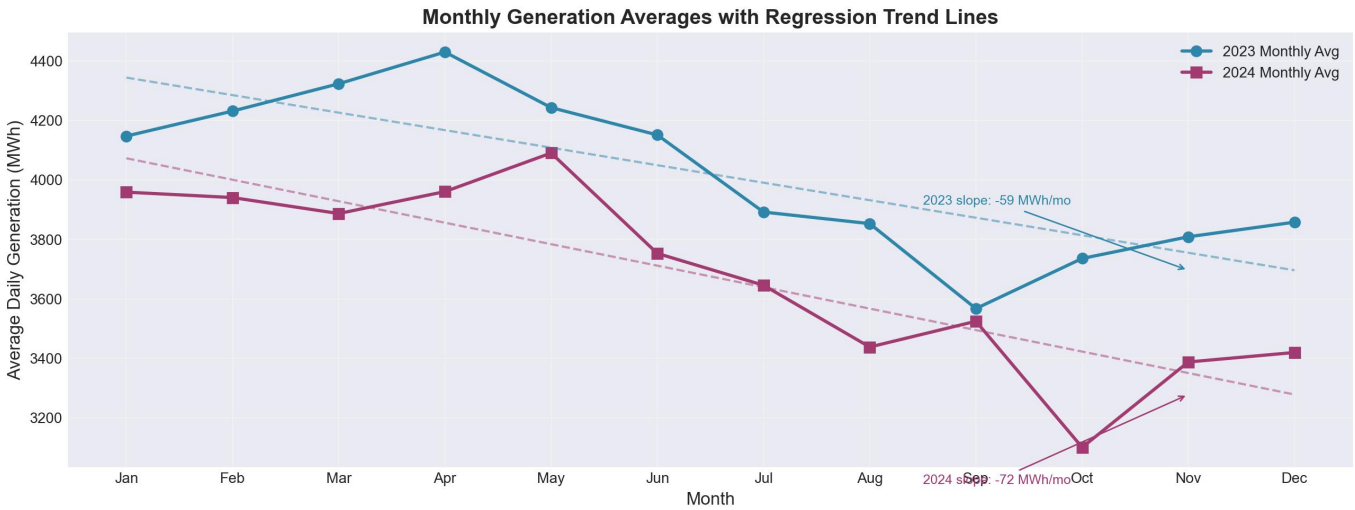

fig_04_monthly_trend.png

Each dot is a monthly average. The dashed lines are the fitted regression trends. The 2024 line starts in January already below where 2023 started, and its slope is steeper throughout the year.

The slope annotations at the end of each trend line show the numbers directly. The 2024 trend drops -72 MWh every month on average versus -59 for 2023. That 13 MWh/month difference compounds. By December, it translates to an additional 156 MWh monthly shortfall relative to 2023's own rate of decline.

April 2023 was the best month in the entire two-year dataset at 4,430 MWh average. October 2024 was the worst at 3,101 MWh. That is a 1,329 MWh swing between the peak and the floor, across two years of the same grid.

The Seasonal Picture

Power generation follows seasonal patterns driven by weather, demand cycles, and operational rhythms. We need to separate those natural patterns from structural deterioration. Breaking data into quarters does that clearly.

Figure 5: Quarterly Comparison

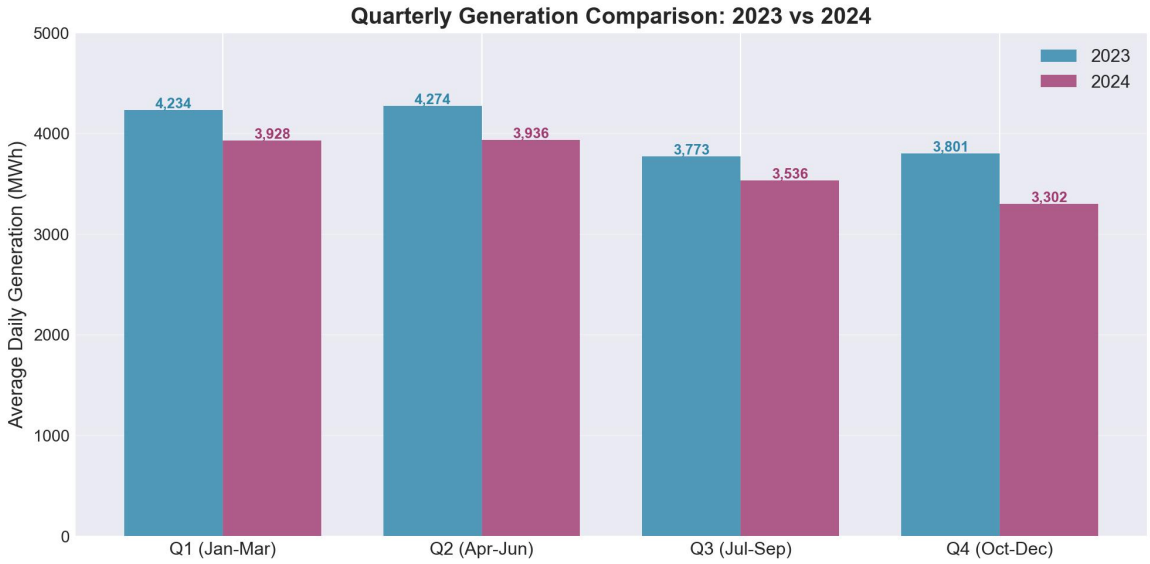

fig_05_quarterly_comparison.png

This chart strips away daily noise entirely and shows one number per quarter per year. The comparison is direct. Blue bars are 2023; pink bars are 2024. The values sit above each bar for easy reading.

Every single quarter in 2024 underperforms its 2023 equivalent:

Quarter	2023	2024	Difference
Q1 (Jan-Mar)	4,234 MWh	3,928 MWh	-306 MWh
Q2 (Apr-Jun)	4,274 MWh	3,936 MWh	-338 MWh
Q3 (Jul-Sep)	3,773 MWh	3,536 MWh	-237 MWh
Q4 (Oct-Dec)	3,801 MWh	3,302 MWh	-499 MWh

Q4 2024 is where the story turns serious. A 499 MWh quarterly drop. In 2023, Q4 was a slight recovery from Q3. In 2024, Q4 fell further. This is where the system stopped merely declining and started deteriorating sharply.

The Grid Stability Problem

Lower generation is one problem. Unstable generation is a different and compounding problem. An unstable grid damages equipment, triggers protective shutdowns, and makes full collapses more likely.

Grid frequency is the most sensitive real-time indicator of grid health. When generation falls short of demand, frequency drops below 50 Hz. When generation exceeds demand, it rises above 50 Hz. The acceptable operating band is 49.75 to 50.25 Hz. Outside that range, equipment stress increases rapidly.

Figure 6: Grid Frequency Over Time

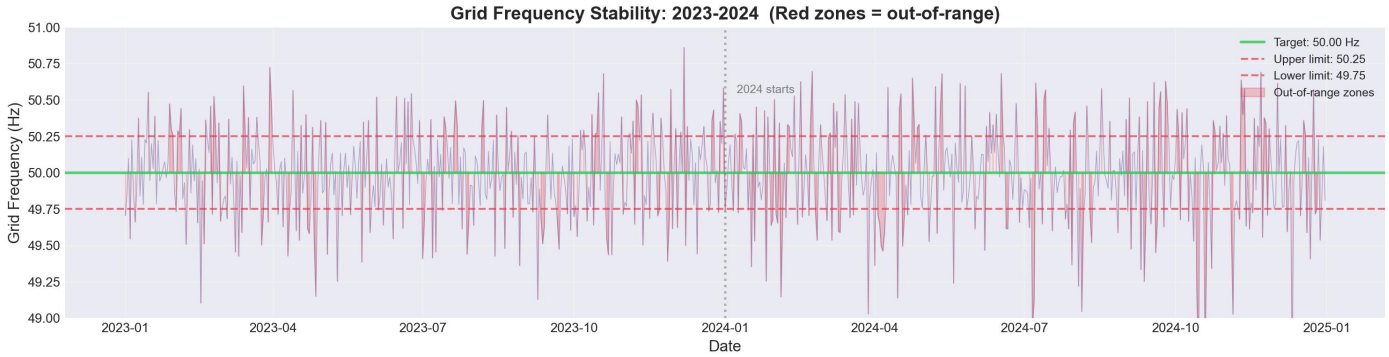

fig_06_grid_frequency.png

The purple line is grid frequency over the full two-year period. The green line is the target at 50 Hz. The red dashed lines mark the upper and lower limits of the safe operating range. The red-shaded areas are periods when frequency was outside those limits.

In 2023, frequency violations occurred on 156 days, or 42.7% of the year. In 2024, on 179 days, or 48.9%. Nearly half of 2024's days saw the grid operating outside safe limits.

That is not occasional instability. That is a system under chronic stress. And the consequence matters: equipment operating outside the safe frequency range ages faster, which creates more component failures, which causes more frequency deviations. The cycle feeds itself, which is why frequency violations are a leading indicator of grid collapses rather than just a symptom of them.

The Anomaly Pattern

Beyond the trend and the variability, we looked for extreme events. Days where generation was so far from normal that they qualified as statistical outliers. We flagged any day where the z-score exceeded 2.5, meaning daily output was more than 2.5 standard deviations from the dataset mean.

- 2023: 2 anomalous days
- 2024: 11 anomalous days

That is a 5.5x increase in catastrophic generation failures from one year to the next.

Figure 7: Anomalous Days on the Time Series

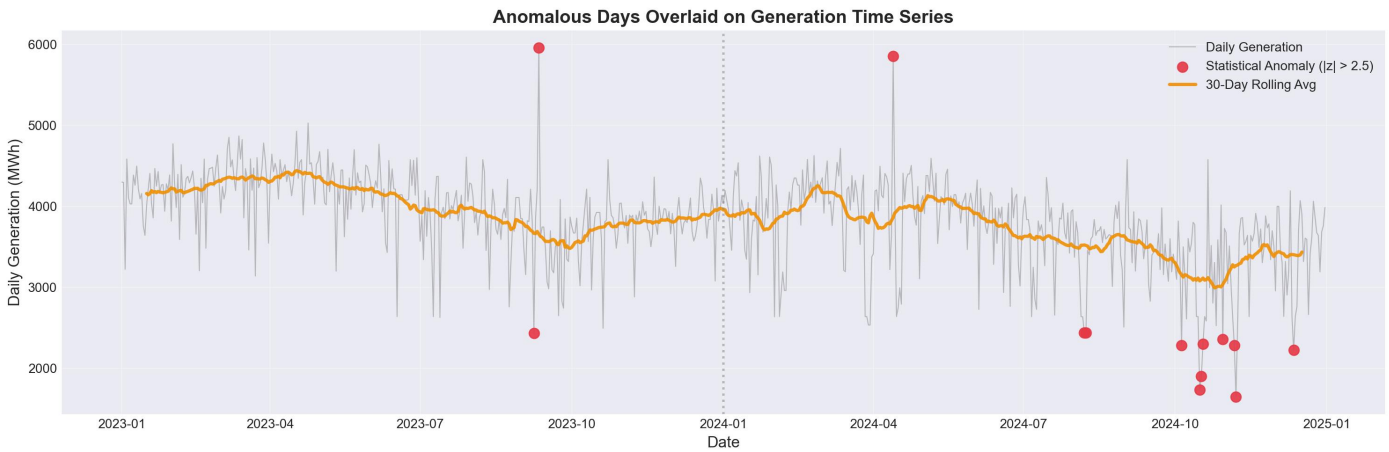

fig_07_anomalies_overlay.png

The red dots mark the anomaly days directly on the time series. The orange line is the 30-day rolling average for trend context. The gray background shows raw daily output.

The clustering tells the story more clearly than any table could. Almost all the red dots appear after July 2024, with the densest cluster in October and November. This is not a random distribution of bad days across two years. It is a concentration of failures in a specific window.

The five worst generation days in the entire dataset:

- November 7, 2024: 1,650 MWh
- October 16, 2024: 1,735 MWh
- October 17, 2024: 1,903 MWh
- December 12, 2024: 2,224 MWh
- November 6, 2024: 2,286 MWh

All five happened in 2024. Four of them in Q4. On those days, the grid delivered less than half of what it typically provides. This is the kind of day where hospitals switch to backup power, where cold storage facilities lose inventory, where the cost goes beyond inconvenience into damage.

The data cannot tell us what caused these specific failures. But it can show us when the system crossed a threshold from chronic under-performance into acute failure events.

The Compounding Cost

Each bad day stacks on top of the previous ones. The total energy cost across a year is visible in cumulative generation.

Figure 8: Cumulative Generation and the Growing Deficit

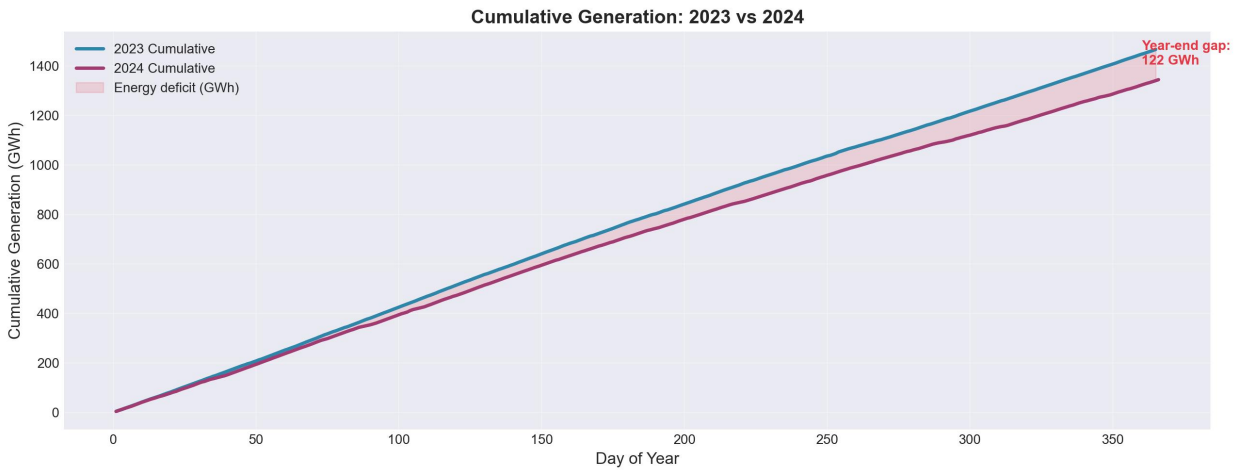

fig_08_cumulative_generation.png

The blue line traces 2023's cumulative generation as days accumulate through the year. The pink line does the same for 2024. The red-shaded area between them is the energy that was not delivered in 2024 relative to the previous year.

That gap starts small in January 2024 and widens every month. The shading grows faster in H2 2024, consistent with the accelerating trend and anomaly clustering we identified earlier.

The year-end gap between the two cumulative lines represents hundreds of GWh of energy that was demanded but not provided. That is not an abstraction. It is the diesel generators that ran instead, the factories that reduced

shifts, the households that went without. The cumulative chart makes visible what the daily average hides: the cost accumulates.

The Full Distribution Picture

One more view. Box plots compress a full year of data into a compact summary: median, interquartile range, whiskers showing the full spread, and outlier points plotted individually.

Figure 9: Box Plot Comparison

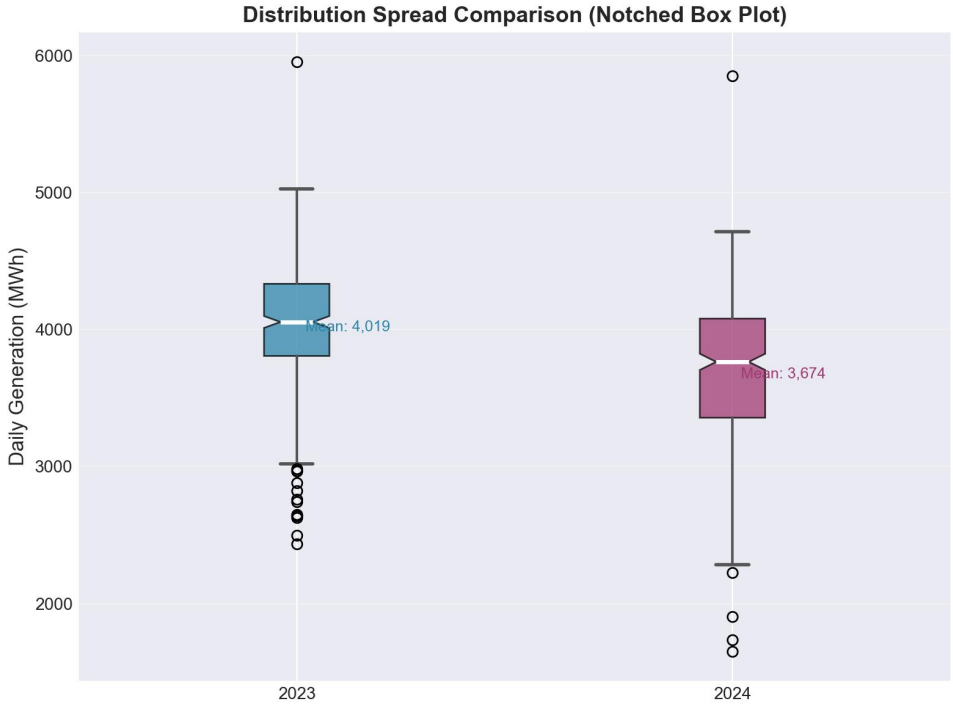

fig_09_boxplot_comparison.png

The 2024 box is taller than the 2023 box. A taller box means a larger interquartile range, which means the middle 50% of days spanned a wider range in 2024. The system was less consistent.

The notches around each median represent confidence intervals on the median value. These notches do not overlap, which is a visual confirmation that the two years have statistically different medians, independent of the t-test on means.

The outlier dots below each box are the extreme low-generation events. There are more of them in 2024, and they sit lower. This view ties together everything we have shown: lower central values, wider spread, and more extreme failures.

Why Did This Happens

This is the honest part.

The data tells us what happened. It does not tell us why. We can observe the decline, the instability, the Q4 deterioration, and the clustering of failures. We cannot, from this dataset alone, assign causes. That would require knowing whether there were gas supply shortages, specific plant failures, maintenance funding gaps, or transmission infrastructure collapses in particular zones.

These are operational questions that require operational data. What we have established is that whatever the cause, the effect was real, large, statistically certain, and accelerating.

One thing worth addressing: the generation-demand gap actually narrowed slightly, from 31.1% in 2023 to 29.8% in 2024. On the surface, this looks like progress. It is probably not. When power supply becomes chronically unreliable, businesses and households adjust their behavior. They scale back activity, shift to generators for primary power, or shut down operations that cannot tolerate outages. Recorded demand falls not because the need decreased but because the expectation of supply has been abandoned. A narrower gap driven by falling effective demand is not improvement. It is the economy adapting to a broken infrastructure.

What Does This Mean

For a household in Lagos, Abuja, or anywhere in between, the 41% increase in variability is particularly difficult. You can build a routine around consistent scarcity. You cannot plan around randomness. When the grid might give you six hours or sixteen hours on any given day with no pattern you can predict, every decision that depends on power becomes uncertain.

For businesses, the doubling of severe-gap days and the 5.5x increase in anomalous days means backup power is no longer a contingency. It is the primary power source, with the grid as occasional supplement. That is an enormous operating cost imposed on every business that cannot afford to stop.

For infrastructure planners, the steeper monthly trend in 2024 ($R^2 = 0.74$) is the most urgent signal. A high R-squared on a downward trend means the decline is systematic, not random. Systematic declines respond to systematic interventions, which means this is a solvable problem. But solving it requires acknowledging the trajectory clearly, which is what this analysis attempts to do.

What Should Be Done Next

Based on what the data shows, the clearest priorities are:

Investigate the H2 2024 inflection point. The anomaly clustering, the Q4 collapse, and the steepening of the rolling average all converge on the second half of 2024 as the moment something changed significantly. Understanding that specific change is the most important analytical next step.

Target frequency stability as a concrete metric. Frequency violations on 49% of days is not a background condition. It is measurable, and reducing it to below 25% within twelve months is a concrete, trackable objective that would reduce the rate at which the system generates its own secondary failures.

Treat the generation-demand gap with more care. If demand measurements are suppressed by chronic under-supply, the gap data understates the true deficit. Better demand estimation is needed to understand how much capacity the country actually needs versus how much it has settled for.

Make generation data public and real-time. This analysis was possible because we had daily data. Decisions affecting 224 million people should be made on the basis of transparent, current data. A system you can measure is one you can manage. A system you cannot measure manages itself, and this analysis shows where that leads.

Closing

The national conversation about electricity in Nigeria is full of opinions and frustrations. What it often lacks is evidence.

This analysis provides some of that. Not all of it. Not with certainty about causes or predictions about what comes next. But with clear answers to clear questions.

Was 2024 worse than 2023? Yes, by 8.6% in average generation and significantly more in reliability.

Was that difference statistically real? Yes, with a p-value of essentially zero and a large effect size.

How fast was the decline? Faster than the year before, with an accelerating monthly slope.

Where did it get worst? Q4 2024, specifically October and November, where the worst five days in the two-year dataset all occurred.

The lights going out more often is not a feeling. It is measurable. And if we can measure it clearly, we can at least start having an honest conversation about what to do.

Appendix A: Statistical Reference

Test or Metric	Value
2023 Mean Generation	4,018.6 MWh
2024 Mean Generation	3,674.4 MWh
Absolute change	-344.2 MWh
Percentage change	-8.57%
t-statistic	8.9469
p-value	< 0.000001
Cohen's d (effect size)	0.6618 (large)
95% CI 2023	[3,972.1, 4,065.1] MWh
95% CI 2024	[3,614.7, 3,734.0] MWh
2023 coefficient of variation	11.2%
2024 coefficient of variation	15.8%
2023 monthly slope	-58.9 MWh/month ($R^2 = 0.63$)
2024 monthly slope	-72.2 MWh/month ($R^2 = 0.74$)
2023 frequency violations	156 days (42.7%)
2024 frequency violations	179 days (48.9%)
2023 anomalous days	2
2024 anomalous days	11

Appendix B: Figure Index

All figures were produced inside the Analysis Notebook at the moment the corresponding analysis was completed.

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